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21 Author

AH Kibbe.

22 Date of Revision

10 February 2009.



Acesulfame Potassium

1 Nonproprietary Names

BP: Acesulfame Potassium
 PhEur: Acesulfame Potassium
 USP-NF: Acesulfame Potassium

2 Synonyms

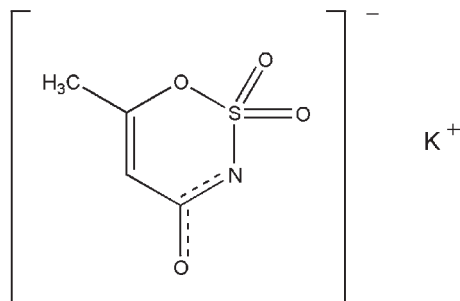
Acesulfame K; ace K; acesulfamum kalicum; E950; 6-methyl-3,4-dihydro-1,2,3-oxathiazin-4(3H)-one-2,2-dioxide potassium salt; potassium 6-methyl-2,2-dioxo-oxathiazin-4-olate; *Sunett*; *Sweet One*.

3 Chemical Name and CAS Registry Number

6-Methyl-1,2,3-oxathiazin-4(3H)-one-2,2-dioxide potassium salt [55589-62-3]

4 Empirical Formula and Molecular Weight

C₄H₄KNO₄S 201.24

5 Structural Formula**6 Functional Category**

Sweetening agent.

7 Applications in Pharmaceutical Formulation or Technology

Acesulfame potassium is used as an intense sweetening agent in cosmetics, foods, beverage products, table-top sweeteners, vitamin and pharmaceutical preparations, including powder mixes, tablets, and liquid products. It is widely used as a sugar substitute in compounded formulations,⁽¹⁾ and as a toothpaste sweetener.⁽²⁾

The approximate sweetening power is 180–200 times that of sucrose, similar to aspartame, about one-third as sweet as sucralose, one-half as sweet as sodium saccharin, and about 4–5 times sweeter

than sodium cyclamate.⁽³⁾ It enhances flavor systems and can be used to mask some unpleasant taste characteristics.

8 Description

Acesulfame potassium occurs as a colorless to white-colored, odorless, crystalline powder with an intensely sweet taste.

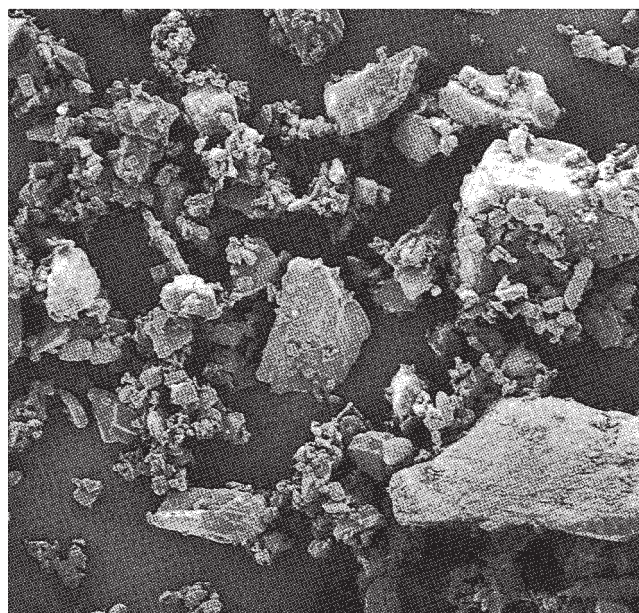
9 Pharmacopeial Specifications

See Table I.

10 Typical Properties

Acidity/alkalinity pH = 5.5–7.5 (1% w/v aqueous solution)
Bonding index 0.007⁽⁴⁾
Brittle fracture index 0.08⁽⁴⁾
Density (bulk) 1.04 g/cm³ ⁽⁴⁾
Density (tapped) 1.28 g/cm³ ⁽⁴⁾
Elastic modulus 4000 MPa⁽⁴⁾
Flowability 19% (Carr compressibility index)⁽⁴⁾
Melting point 250°C
NIR spectra see Figure 1.
Solubility see Table II.

SEM 1: Excipient: acesulfame potassium; magnification: 150×; voltage: 5 kV.



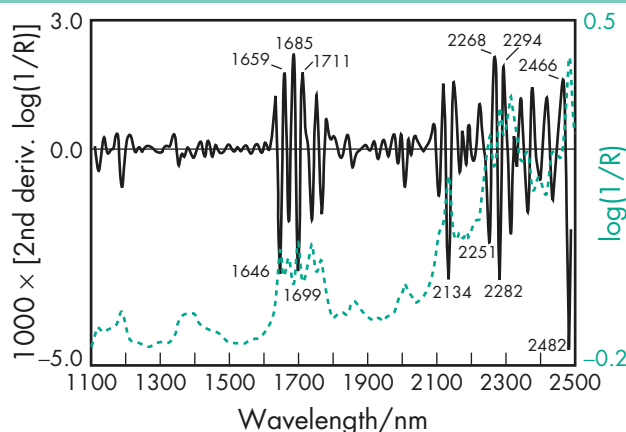


Figure 1: Near-infrared spectrum of acesulfame potassium measured by reflectance.

Table I: Pharmacopeial specifications for acesulfame potassium.

Test	PhEur 6.0	USP32-NF27
Characters	+	—
Identification	+	+
Appearance of solution	+	—
Acidity or alkalinity	+	+
Acetylacetamide	0.125%	—
Impurity B ^(a)	≤20 ppm	—
Unspecified impurities	≤0.1%	≤0.002%
Total impurities	≤0.1%	—
Fluorides	≤3 ppm	≤0.0003%
Heavy metals	≤5 ppm	≤10 µg/g
Loss on drying	≤1.0%	≤1.0%
Assay (dried basis)	99.0–101.0%	99.0–101.0%

(a) Impurity B is 5-chloro-6-methyl-1,2,3-oxathiazin-4(3H)-one 2,2-dioxide.

Table II: Solubility of acesulfame potassium.⁽³⁾

Solvent	Solubility at 20°C unless otherwise stated
Ethanol	1 in 1000
Ethanol (50%)	1 in 10
Ethanol (15%)	1 in 4.5
Water	1 in 6.7 at 0°C 1 in 3.7 at 20°C 1 in 0.77 at 100°C

Specific volume 0.538 cm³/g⁽⁵⁾

Surface tension 73.2 mN/m⁽⁶⁾ (1% w/v aqueous solution at 20°C)

Tensile strength 0.5 MPa⁽⁴⁾

Viscoelastic index 2.6⁽⁴⁾

11 Stability and Storage Conditions

Acesulfame potassium possesses good stability. In the bulk form it shows no sign of decomposition at ambient temperature over many years. In aqueous solutions (pH 3.0–3.5 at 20°C) no reduction in sweetness was observed over a period of approximately 2 years. Stability at elevated temperatures is good, although some decomposition was noted following storage at 40°C for several months. Sterilization and pasteurization do not affect the taste of acesulfame potassium.⁽⁷⁾

The bulk material should be stored in a well-closed container in a cool, dry place and protected from light.

12 Incompatibilities

13 Method of Manufacture

Acesulfame potassium is synthesized from acetoacetic acid *tert*-butyl ester and fluorosulfonyl isocyanate. The resulting compound is transformed to fluorosulfonyl acetoacetic acid amide, which is then cyclized in the presence of potassium hydroxide to form the oxathiazinone dioxide ring system. Because of the strong acidity of this compound, the potassium salt is produced directly.⁽⁸⁾

An alternative synthesis route for acesulfame potassium starts with the reaction between diketene and amidosulfonic acid. In the presence of dehydrating agents, and after neutralization with potassium hydroxide, acesulfame potassium is formed.

14 Safety

Acesulfame potassium is widely used in beverages, cosmetics, foods, and pharmaceutical formulations, and is generally regarded as a relatively nontoxic and nonirritant material. Pharmacokinetic studies have shown that acesulfame potassium is not metabolized and is rapidly excreted unchanged in the urine. Long-term feeding studies in rats and dogs showed no evidence to suggest acesulfame potassium is mutagenic or carcinogenic.⁽⁹⁾

The WHO has set an acceptable daily intake for acesulfame potassium of up to 15 mg/kg body-weight.⁽⁹⁾ The Scientific Committee for Foods of the European Union has set a daily intake value of up to 9 mg/kg of body-weight.⁽³⁾

LD₅₀ (rat, IP): 2.2 g/kg⁽⁷⁾

LD₅₀ (rat, oral): 6.9–8.0 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection, gloves, and a dust mask are recommended.

16 Regulatory Status

Included in the FDA Inactive Ingredients Database for oral and sublingual preparations. Included in the Canadian List of Acceptable Non-medicinal Ingredients. Accepted for use in Europe as a food additive. It is also accepted for use in certain food products in the USA and several countries in Central and South America, the Middle East, Africa, Asia, and Australia.

17 Related Substances

Alitame.

18 Comments

The perceived intensity of sweeteners relative to sucrose depends upon their concentration, temperature of tasting, and pH, and on the flavor and texture of the product concerned.

Intense sweetening agents will not replace the bulk, textural, or preservative characteristics of sugar, if sugar is removed from a formulation.

Synergistic effects for combinations of sweeteners have been reported, e.g. acesulfame potassium with aspartame or sodium cyclamate; *see also* Aspartame. A ternary combination of sweeteners that includes acesulfame potassium and sodium saccharin has a greater decrease in sweetness upon repeated tasting than other combinations.⁽¹⁰⁾

Note that free acesulfame acid is not suitable for use as a sweetener.

A specification for acesulfame potassium is contained in the Food Chemicals Codex (FCC).⁽¹¹⁾

19 Specific References

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- 3 Wilson R, ed. *Sweeteners*, 3rd edn. Oxford, UK: Blackwell Publishing, 2007; 3–19.
- 4 Mullarney MP *et al.* The powder flow and compact mechanical properties of sucrose and three high-intensity sweeteners used in chewable tablets. *Int J Pharm* 2003; 257: 227–236.
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- 7 Lipinski G-WvR, Huddart BE. Acesulfame K. *Chem Ind* 1983; 11: 427–432.
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- 9 FAO/WHO. Evaluation of certain food additives and contaminants. Thirty-seventh report of the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 1991; No. 806.
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20 General References

- Anonymous. Artificial sweeteners. *Can Pharm J* 1996; 129: 22.
 Lipinski G-WvR, Lück E. Acesulfame K: a new sweetener for oral cosmetics. *Manuf Chem* 1981; 52(5): 37.
 Marie S. Sweeteners. Smith J, ed. *Food Additives User's Handbook*. Glasgow: Blackie, 1991; 47–74.
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21 Author

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22 Date of Revision

26 February 2009.

Acetic Acid, Glacial

1 Nonproprietary Names

BP: Glacial Acetic Acid

JP: Glacial Acetic Acid

PhEur: Acetic Acid, Glacial

USP: Glacial Acetic Acid

2 Synonyms

Acidum aceticum glaciale; E260; ethanoic acid; ethylic acid; methane carboxylic acid; vinegar acid.

See also Sections 17 and 18.

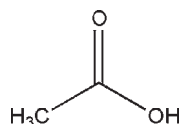
3 Chemical Name and CAS Registry Number

Ethanoic acid [64-19-7]

4 Empirical Formula and Molecular Weight

C₂H₄O₂ 60.05

5 Structural Formula



6 Functional Category

Acidifying agent.

7 Applications in Pharmaceutical Formulations or Technology

Glacial and diluted acetic acid solutions are widely used as acidifying agents in a variety of pharmaceutical formulations and food preparations. Acetic acid is used in pharmaceutical products as a buffer system when combined with an acetate salt such as sodium

acetate. Acetic acid is also claimed to have some antibacterial and antifungal properties.

8 Description

Glacial acetic acid occurs as a crystalline mass or a clear, colorless volatile solution with a pungent odor.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for glacial acetic acid.

Test	JP XV	PhEur 6.0	USP 32
Identification	+	+	+
Characters	+	+	—
Freezing point	≥ 14.5°C	≥ 14.8°C	≥ 15.6°C
Nonvolatile matter	≤ 1.0 mg	≤ 0.01%	≤ 1.0 mg
Sulfate	+	+	+
Chloride	+	+	+
Heavy metals	≤ 10 ppm	≤ 5 ppm	≤ 5 ppm
Iron	—	≤ 5 ppm	—
Readily oxidizable impurities	+	+	+
Assay	≥ 99.0%	99.5–100.5%	99.5–100.5%

10 Typical Properties

Acidity/alkalinity

pH = 2.4 (1 M aqueous solution);

pH = 2.9 (0.1 M aqueous solution);

pH = 3.4 (0.01 M aqueous solution).

Boiling point 118°C

Dissociation constant pK_a = 4.76

Flash point 39°C (closed cup); 57°C (open cup).

Melting point 17°C

Refractive index n_D²⁰ = 1.3718